DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 11: Beta and the Security Market Line

Interpreting β · The SML · Alpha · Portfolio Beta

August 13, 2026

Today's Agenda

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3. The SML, revisited
4. Alpha: mispricing relative to CAPM

Learning Objectives

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- Define alpha as a mispricing signal.

Beta: A Volume Knob for Market Moves

Speakers Tuned to the Market's Signal

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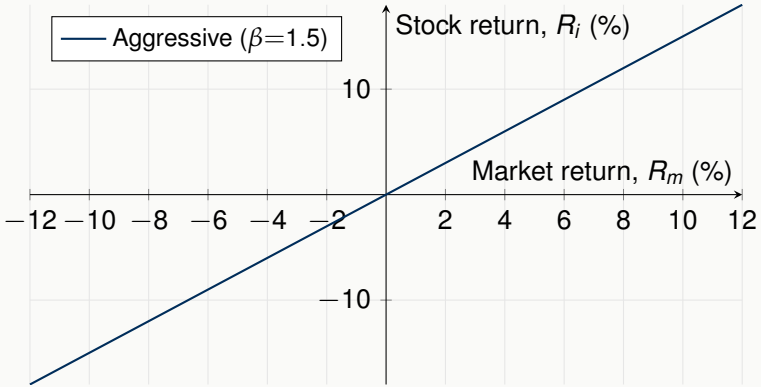
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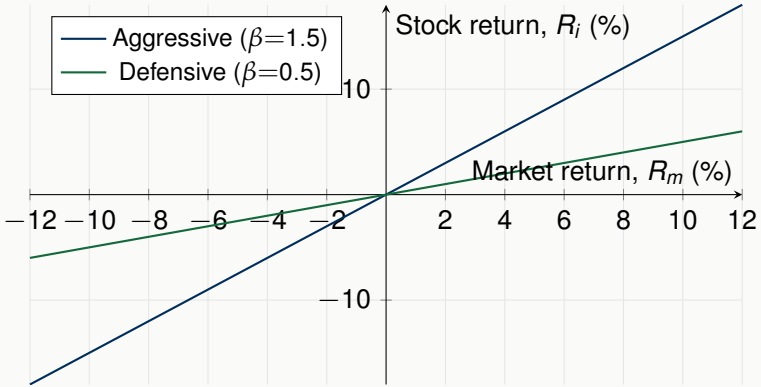
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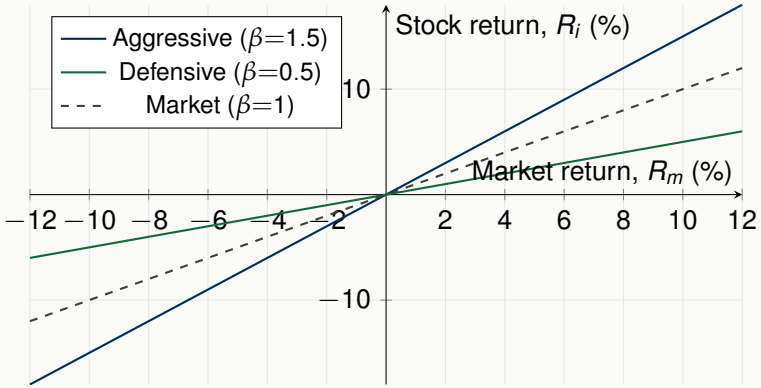
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- $\beta < 0$: **rare** but real – moves opposite the market

Portfolio Beta: A Weighted Average

Formula

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$$r_f = 5\%, \mathbb{E}[R_m] = 11.25\% \Rightarrow \mathbb{E}[R_p] = 5 + 1.135(6.25) = 12.09\%$$

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$$\alpha_i = \mathbb{E}[R_i]_{\text{actual}} - \left(r_f + \beta_i(\mathbb{E}[R_m] - r_f)\right)$$

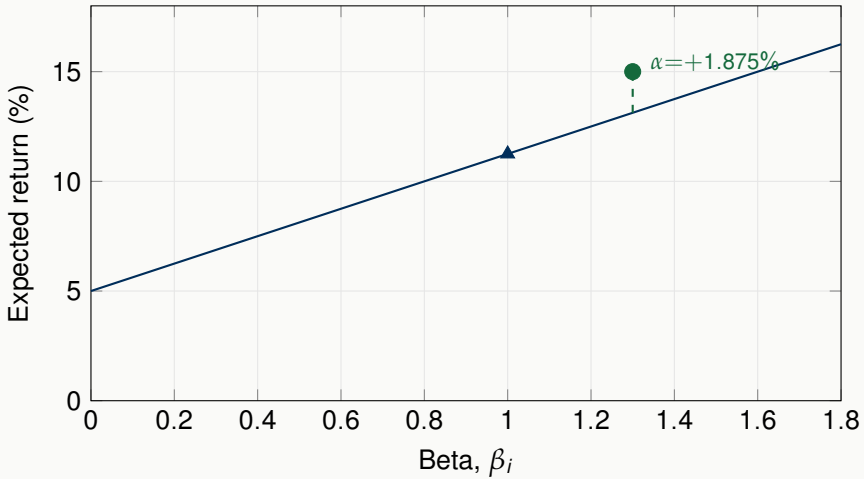
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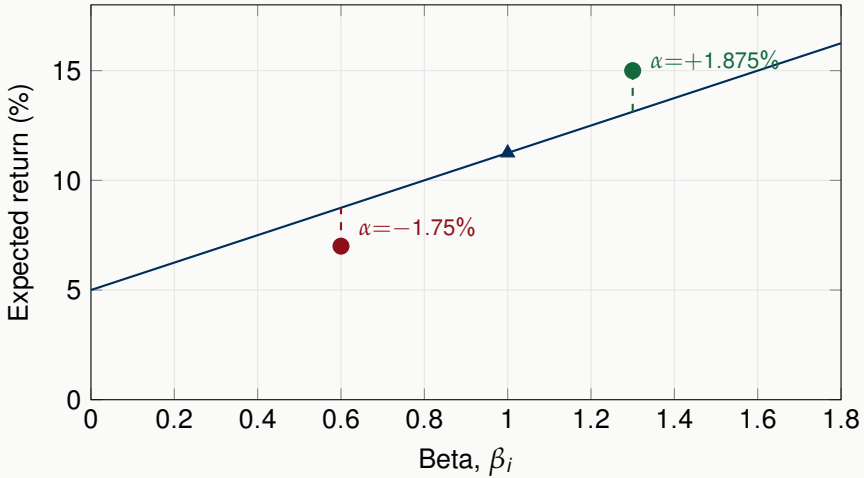
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The vertical distance between actual and CAPM-predicted expected return.

Alpha on the SML



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Alpha as a Report Card

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The SML is the “expected” grade purely from hours studied (beta). Scoring *above* that – skill, luck, insider knowledge – is positive alpha. Scoring *below* is negative alpha. It says nothing about whether outperformance will *persist*.

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- $\beta > 1$ aggressive, $0 < \beta < 1$ defensive, $\beta = 0$ neutral, $\beta < 0$ rare/hedging.
- Portfolio beta $= \sum_i w_i \beta_i$.
- **Alpha** measures the gap between actual and CAPM-predicted return – mispricing or an omitted factor.

Check Your Understanding

Discuss (2 minutes)

1. Four assets, equal weights, betas 1.6, 0.9, 0.3, 1.2. Find β_p.

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2. $r_f = 6\%$, $\mathbb{E}[R_m] = 13\%$, $\beta = 0.8$, actual $\mathbb{E}[R_i] = 12\%$. Find alpha.
3. Why might a high raw return with alpha near zero not indicate skill?

Looking Ahead

Everything so far assumed we already *know* beta.

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Estimating Beta from Data – Lecture 12